One Gap, Three Answers:
A Calculus Analysis of Receptive and Expressive Language Development

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I began this project by looking at how two parts of language development change over early childhood. Receptive language concerns what a child understands, while expressive language concerns what the child can communicate. I wanted to compare two genetic neurodevelopmental conditions in which those abilities can develop unevenly and for which comparable developmental data were available. Down syndrome (DS) and Fragile X syndrome (FXS) fit both requirements. DS is caused by an extra copy of chromosome 21. FXS has a different genetic cause: a short section of DNA in the *FMR1* gene is repeated too many times. When the repeat becomes very large, the gene is largely switched off, reducing production of a protein important for nervous-system development (MedlinePlus Genetics, n.d.; National Institute of Child Health and Human Development, n.d.). In a recently published study, D'Souza et al. (2026) reported receptive language as a relative strength and expressive language as a relative weakness in young children with DS, while Roberts et al. (2009) found significantly faster receptive than expressive language growth in boys with FXS.

Both source studies measured early development with the Mullen Scales of Early Learning (MSEL), which assesses Gross Motor, Fine Motor, Visual Reception, Receptive Language, and Expressive Language in children from birth through 68 months (Pearson Assessments, n.d.). This project uses the receptive- and expressive-language results reported in those studies. The studies express these results as age-equivalent (AE) scores in months. An AE score of 24 means the raw subscale score corresponds approximately to the score associated with 24 months of age in the normative sample. It does not mean every aspect of a child's language is literally at a 24-month developmental level. Because AE scores are referenced to age norms, chronological age provides the typical-development baseline used later in the models (D'Souza et al., 2026; Roberts et al., 2009).

My original question was how the difference between receptive and expressive language changes over time in DS and FXS. I first treated that difference as a developmental gap and defined it as $G(t) = R(t) - E(t)$. Once I began working with the fitted trajectories, however, the project changed. The vertical difference between the curves was only one reasonable way to measure their separation. The same trajectories could also be compared proportionally or by the chronological time separating them from reaching the same numerical developmental score. That led to a more fundamental question: if the same developmental relationship can be measured in several valid ways, what exactly should "the gap" mean?

Problem. Using linear models based on the published DS and FXS data, (a) use derivatives to determine how the absolute and proportional receptive-expressive gaps change with age, (b) find a general relationship between the time gap and the score gap, (c) compute all three measures at an AE score of 24 for each group, and (d) find the ages at which the score-gap and time-gap comparisons between DS and FXS switch. The rest of this paper works through the solution.

Developmental Models and Methods

D'Souza et al. (2026) reported linear receptive- and expressive-language trajectories for children with DS. Roberts et al. (2009) reported MSEL age-group means for boys with FXS, which I modeled with linear functions to approximate the overall receptive- and expressive-language trends. Let t be chronological age in months, $R(t)$ the receptive-language AE score, and $E(t)$ the expressive-language AE score. The models are:

Down syndrome	$R_{DS}(t) = 0.48845t + 2.176$	$E_{DS}(t) = 0.33714t + 4.904$
Fragile X	$R_{FXS}(t) = 0.44977t + 3.30959$	$E_{FXS}(t) = 0.39539t + 3.23195$

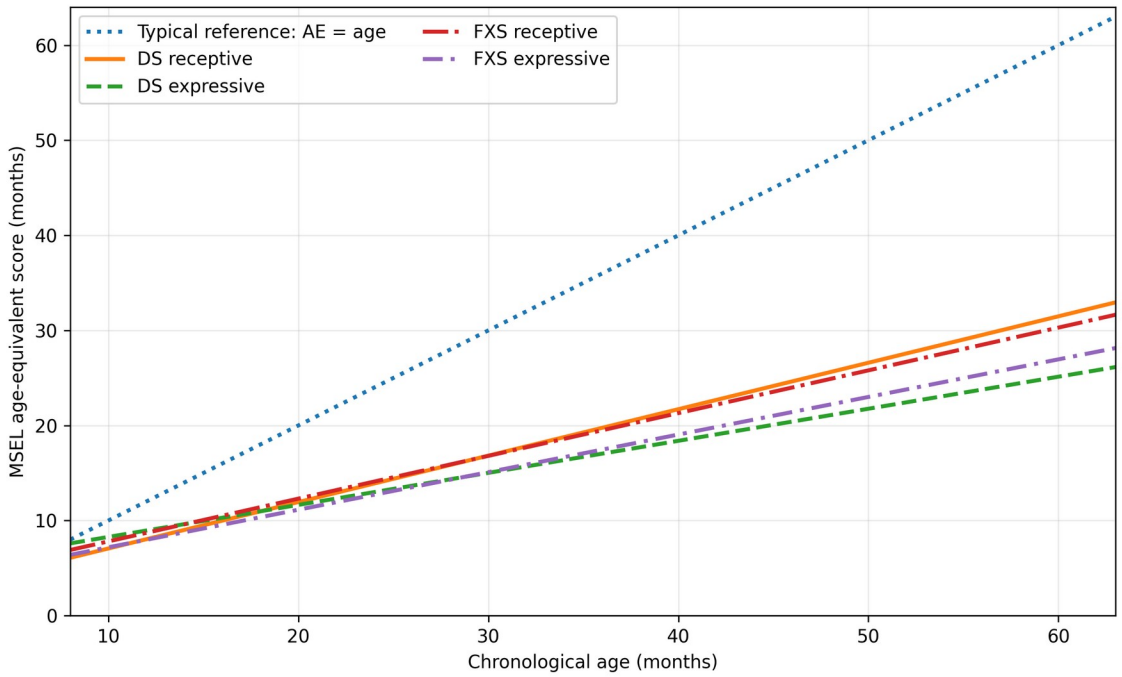
For comparison, I normalize typical development so that the expected age-equivalent score equals chronological age, consistent with the age-referenced interpretation of AE scores used in both source studies (D'Souza et al., 2026; Roberts et al., 2009):

Typical reference $R_{TD}(t) = E_{TD}(t) = t$

The DS equations are approximate because D'Souza et al. (2026) did not report the slope and intercept values directly. I estimated the slopes from the linear trajectories shown in Figure 2 of D'Souza et al. (2026) and determined the intercepts from the group means reported in Table 2 of the same study, which correspond to a mean chronological age of 27.81 months. For FXS, I used ordinary least squares to construct linear functions from the seven age-cluster means reported in Table II of Roberts et al. (2009), using the nominal cluster ages as the t-values. Because the clusters cover age ranges rather than single exact ages, these equations approximate the published group trends and are not reproductions of the authors' hierarchical linear models. The FXS equations above are rounded, and the FXS calculations that follow use the unrounded regression coefficients.

Figure 1

Receptive and Expressive MSEL Age-Equivalent Trajectories



Note. The dotted AE = age line is the normalized typical-development reference. The remaining lines show the DS and FXS receptive and expressive language models used in the analysis. The figure was generated in Python from the equations above; see Appendix A.

Three Ways to Measure the Gap

Absolute Score Gap

The vertical difference at the same chronological age is

$$G(t) = R(t) - E(t)$$

$$G_{DS}(t) = 0.15131t - 2.728 \qquad G_{FXS}(t) = 0.05438t + 0.07763 \qquad G_{TD}(t) = 0$$

Differentiating gives $G'_{DS}(t) = 0.15131$, $G'_{FXS}(t) = 0.05438$, and $G'_{TD}(t) = 0$. Both fitted gap functions increase with age, and the DS gap increases faster. Because $G_{DS}(t) < 0$ before $t \approx 18.03$ months, the DS expressive score is higher at first, so the DS gap changes sign instead of widening the whole time.

Proportional Gap

A raw difference does not show how large the gap is relative to the expressive score. Define

$$Q(t) = R(t) / E(t) = 1 + G(t) / E(t)$$

For linear models $R(t) = m_R t + b_R$ and $E(t) = m_E t + b_E$, the quotient rule gives

$$Q'(t) = [m_R b_E - b_R m_E] / [m_E t + b_E]^2$$

$$Q'_{DS}(t) = 1.66174 / (0.33714t + 4.904)^2 > 0$$

$$Q'_{FXS}(t) = 0.14506 / (0.39539t + 3.23195)^2 > 0$$

Thus the proportional receptive advantage increases in both fitted models wherever the expressive score is positive. The numerator controlling the sign is positive in both models and substantially larger for DS.

Chronological Time Gap

Let A be a numerical AE score, $t_R(A) = R^{-1}(A)$, and $t_E(A) = E^{-1}(A)$. Define

$$H(A) = t_E(A) - t_R(A)$$

This is the chronological time separating the two trajectories reaching the same numerical AE score.

Proposition 1. Let R and E be continuous, differentiable, strictly increasing developmental trajectories on an interval containing the relevant times, with $E'(t) > 0$. For any numerical AE score A in the common range, there is some c between $t_R(A)$ and $t_E(A)$ such that

$$H(A) = G(t_R(A)) / E'(c)$$

Proof. Since $R(t_R(A)) = A = E(t_E(A))$,

$$G(t_R(A)) = E(t_E(A)) - E(t_R(A)).$$

If $t_R(A) = t_E(A)$, both sides of the proposition equal 0 and there is nothing to prove. Otherwise, by the Mean Value Theorem, for some c strictly between those two times,

$$E(t_E(A)) - E(t_R(A)) = E'(c)[t_E(A) - t_R(A)] = E'(c)H(A).$$

Therefore $G(t_R(A)) = E'(c)H(A)$, so $H(A) = G(t_R(A)) / E'(c)$.

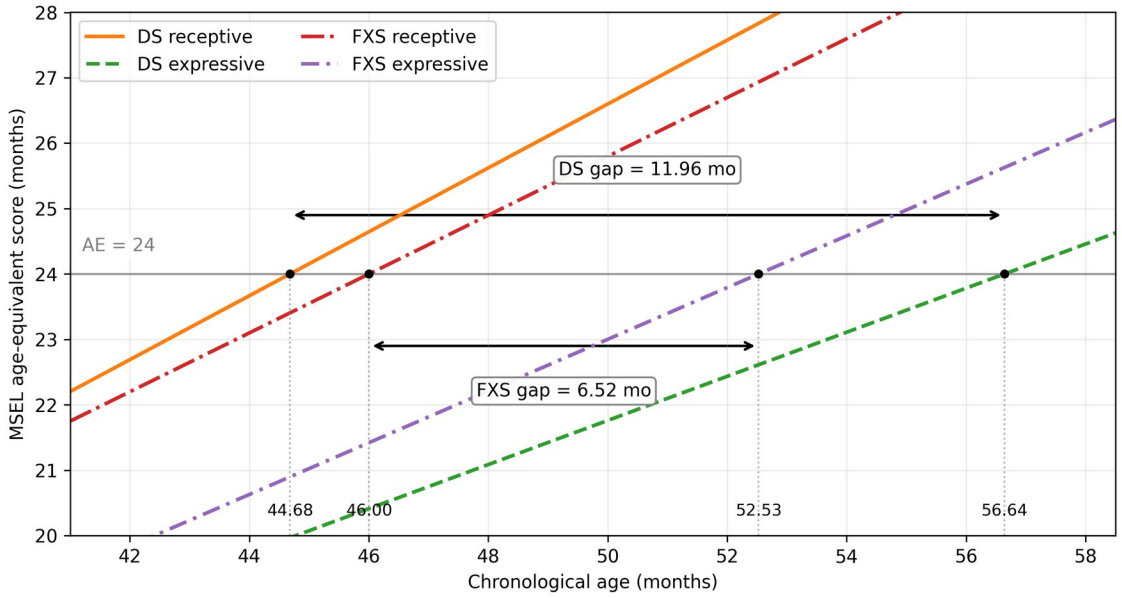
For the linear models in this project, $E'(t) = m_E$ everywhere, so the proposition reduces to $H(A) = G(t_R(A)) / m_E$. The units are AE months divided by AE months per chronological month, leaving chronological months. The result is not limited to straight lines; for a nonlinear trajectory, the Mean Value Theorem gives the same relationship using $E'(c)$ at some point during the interval.

One Relationship, Three Answers

Take $A = 24$, a common AE score reached by both fitted receptive and expressive trajectories within the observed age ranges. For DS, $t_R(24) \approx 44.68$ and $t_E(24) \approx 56.64$, so $H_{DS}(24) \approx 11.96$ chronological months. At $t_R(24)$, $G_{DS} \approx 4.03$ AE months and $Q_{DS} \approx 1.202$. For FXS, $t_R(24) \approx 46.00$ and $t_E(24) \approx 52.53$, so $H_{FXS}(24) \approx 6.52$ months; $G_{FXS} \approx 2.58$ AE months and $Q_{FXS} \approx 1.120$.

Figure 2

Chronological Time Gaps at AE = 24 Months



Note. The horizontal line marks AE = 24 months. The labeled arrows show the chronological time required for each expressive-language trajectory to reach AE = 24 after the corresponding receptive-language trajectory has already reached that value. In the worked example, the DS time gap is 11.96 months and the FXS time gap is 6.52 months. The figure was generated in Python from the equations above; see Appendix B.

Table 1

Three Measures of the Receptive-Expressive Relationship at Receptive AE = 24

Measure	Typical reference	Down syndrome	Fragile X
Chronological age at receptive AE = 24	24.00 mo	44.68 mo	46.00 mo
Absolute score gap, G	0 AE mo	4.03 AE mo	2.58 AE mo
Proportional excess, $Q - 1$	0%	20.2%	12.0%
Chronological time gap, H	0 mo	11.96 mo	6.52 mo

At this point, the DS gap is about 1.56 times the FXS gap by absolute score, 1.68 times as large by proportional excess, and 1.83 times as large by chronological time. All three calculations are correct, but they emphasize different features of the same fitted relationship.

When Correct Measures Disagree

The measures can even reverse the apparent ordering. The absolute gaps are equal when

$$0.15131t - 2.728 = 0.05438t + 0.07763$$

giving $t \approx 28.94$ months.

For a time-based comparison at chronological age t, each group's receptive trajectory has reached $A = R(t)$. The linear form of Proposition 1 therefore gives $H(R(t)) = G(t) / m_E$. Setting the two time gaps equal gives

$$(0.15131t - 2.728) / 0.33714 = (0.05438t + 0.07763) / 0.39539$$

giving $t \approx 26.63$ months.

Between about 26.6 and 28.9 months, the absolute score-gap measure identifies FXS as larger while the time-gap measure identifies DS as larger. Neither result is wrong. The two metrics answer different questions. Because the models come from separate studies and the interval is narrow, this reversal is a mathematical proof of concept, not a clinical ranking of the conditions.

Conclusion

This project began with a basic question about how the receptive-expressive language gap changes over time, but the analysis showed that there is no single mathematical quantity called "the gap." $G(t)$, $Q(t)$, and $H(A)$ are all valid measures of the same two trajectories, yet each preserves different information. The calculus also showed why those measures behave differently: the absolute gap tracks score separation, the proportional gap expresses that separation relative to the expressive score, and the time gap depends on both the score difference and the expressive rate of change. Once the measure is defined, the calculation is unambiguous. The important choice comes earlier, in deciding what kind of difference is actually being measured.

The DS and FXS models show that this choice can materially change the result. At the same developmental target, the relative size of the two modeled gaps changes depending on whether the comparison is made by absolute score, proportion, or chronological time. Over a narrow interval, the absolute and time-based measures even disagree about which condition has the larger separation. The underlying trajectories have not changed. What changed was the mathematical definition applied to them. A correct calculation, therefore, can still give an incomplete picture if the quantity being calculated does not match the question being asked.

That matters because developmental profiles are used to understand children's patterns of strengths and difficulties, not just to produce test scores. D'Souza et al. (2026) caution that easily observed abilities such as expressive language may be used as proxies for overall development in young children with DS, potentially underestimating abilities in other domains and influencing the opportunities offered to them. They therefore emphasize awareness of the uneven developmental profile, individual variation, and continued monitoring. Roberts et al. (2009) similarly argue that understanding developmental profiles in young children with FXS is important for accurate diagnosis, treatment, and support. The mathematics in this project does not determine which measure should be used clinically, but it shows that the way a receptive-expressive difference is represented can make different features of that developmental profile more or less visible.

For language development, that choice has a real consequence. A few age-equivalent months can look like a small difference on paper, but under a slower expressive growth rate the same separation can represent many months of chronological time. Each measure describes a different part of the changing distance between what a child understands and what they can communicate. How the gap is defined decides which part of that relationship becomes visible, so asking the right question of the data matters as much as getting the calculation right.

References

D'Souza, H., Karmiloff-Smith, A., Mareschal, D., & Thomas, M. S. C. (2026). The Down syndrome profile emerges gradually across early development. *Journal of Applied Research in Intellectual Disabilities, 39*(4), Article e70218. https://doi.org/10.1111/jar.70218

MedlinePlus Genetics. (n.d.). *Fragile X syndrome*. U.S. National Library of Medicine. https://medlineplus.gov/genetics/condition/fragile-x-syndrome/

National Institute of Child Health and Human Development. (n.d.). *Down syndrome*. https://www.nichd.nih.gov/health/topics/factsheets/downsyndrome

Pearson Assessments. (n.d.). *Mullen Scales of Early Learning*. https://www.pearsonassessments.com/en-us/Store/Professional-Assessments/Developmental-Early-Childhood/Mullen-Scales-of-Early-Learning/p/100000306

Roberts, J. E., Mankowski, J. B., Sideris, J., Goldman, B. D., Hatton, D. D., Mirrett, P. L., Baranek, G. T., Reznick, J. S., Long, A. C. J., & Bailey, D. B., Jr. (2009). Trajectories and predictors of the development of very young boys with fragile X syndrome. *Journal of Pediatric Psychology, 34*(8), 827–836. https://doi.org/10.1093/jpepsy/jsn129

Appendix A

Python Code Used to Generate Figure 1

The following Python code generates Figure 1 from the developmental equations used in the analysis. It plots the overall receptive- and expressive-language trajectories and the normalized typical-development reference.

```
import numpy as np
import matplotlib.pyplot as plt

# Developmental model coefficients

# Down syndrome
mR_DS = 0.48845
bR_DS = 2.176
mE_DS = 0.33714
bE_DS = 4.904

# Fragile X syndrome (least-squares fit to the age-cluster means
# reported in Table II of Roberts et al., 2009)
cluster_age = np.array([9, 12, 18, 24, 36, 48, 60])
rl_means = np.array([7.00, 8.94, 10.68, 14.53, 19.27, 27.21, 28.64])
el_means = np.array([6.18, 7.76, 9.42, 13.71, 18.17, 24.55, 24.68])
mR_FXS, bR_FXS = np.polyfit(cluster_age, rl_means, 1)
mE_FXS, bE_FXS = np.polyfit(cluster_age, el_means, 1)

# Chronological age range
age = np.linspace(8, 63, 400)

# Typical-development reference
typical = age

# Receptive and expressive trajectories
R_DS = mR_DS * age + bR_DS
E_DS = mE_DS * age + bE_DS

R_FXS = mR_FXS * age + bR_FXS
E_FXS = mE_FXS * age + bE_FXS

# Create figure
fig, ax = plt.subplots(figsize=(8.8, 5.4), dpi=200)

ax.plot(
    age, typical,
    color="tab:blue",
    linestyle=":",
    linewidth=2.2,
    label="Typical reference: AE = age"
)

ax.plot(
    age, R_DS,
    color="tab:orange",
    linestyle="-",
    linewidth=2.2,
    label="DS receptive"
)

ax.plot(
    age, E_DS,
    color="tab:green",
```

```
    linestyle="--",
    linewidth=2.2,
    label="DS expressive"
)

ax.plot(
    age, R_FXS,
    color="tab:red",
    linestyle="-.",
    linewidth=2.2,
    label="FXS receptive"
)

ax.plot(
    age, E_FXS,
    color="tab:purple",
    linestyle=(0, (5, 2, 1, 2)),
    linewidth=2.2,
    label="FXS expressive"
)

ax.set_xlim(8, 63)
ax.set_ylim(0, 64)

ax.set_xlabel("Chronological age (months)")
ax.set_ylabel("MSEL age-equivalent score (months)")

ax.grid(True, alpha=0.25)
ax.legend(loc="upper left", ncol=2, frameon=True)

fig.tight_layout()

fig.savefig(
    "figure1_clean_trajectories.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```

Appendix B

Python Code Used to Generate Figure 2

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.patches import FancyArrowPatch

# Developmental model coefficients

# Down syndrome
mR_DS = 0.48845
bR_DS = 2.176
mE_DS = 0.33714
bE_DS = 4.904

# Fragile X syndrome (least-squares fit to the age-cluster means
# reported in Table II of Roberts et al., 2009)
cluster_age = np.array([9, 12, 18, 24, 36, 48, 60])
rl_means = np.array([7.00, 8.94, 10.68, 14.53, 19.27, 27.21, 28.64])
el_means = np.array([6.18, 7.76, 9.42, 13.71, 18.17, 24.55, 24.68])
mR_FXS, bR_FXS = np.polyfit(cluster_age, rl_means, 1)
mE_FXS, bE_FXS = np.polyfit(cluster_age, el_means, 1)

# Worked example
A = 24

# Chronological ages at which each trajectory reaches AE = 24
tR_DS = (A - bR_DS) / mR_DS
tE_DS = (A - bE_DS) / mE_DS

tR_FXS = (A - bR_FXS) / mR_FXS
tE_FXS = (A - bE_FXS) / mE_FXS

# Chronological time gaps
H_DS = tE_DS - tR_DS
H_FXS = tE_FXS - tR_FXS

# Focused age range for the worked example
age = np.linspace(41, 58.5, 300)

R_DS = mR_DS * age + bR_DS
E_DS = mE_DS * age + bE_DS

R_FXS = mR_FXS * age + bR_FXS
E_FXS = mE_FXS * age + bE_FXS

# Create figure
fig, ax = plt.subplots(figsize=(8.8, 4.8), dpi=200)

ax.plot(
    age, R_DS,
    color="tab:orange",
    linestyle="-",
    linewidth=2.0,
    label="DS receptive"
)

ax.plot(
    age, E_DS,
    color="tab:green",
    linestyle="--",
    linewidth=2.0,
```

```
    label="DS expressive"
)

ax.plot(
    age, R_FXS,
    color="tab:red",
    linestyle="-.",
    linewidth=2.0,
    label="FXS receptive"
)

ax.plot(
    age, E_FXS,
    color="tab:purple",
    linestyle=(0, (5, 2, 1, 2)),
    linewidth=2.0,
    label="FXS expressive"
)

# Horizontal reference line at AE = 24
ax.axhline(
    A,
    color="gray",
    linewidth=1.2,
    alpha=0.75
)

ax.text(
    41.2, 24.35,
    "AE = 24",
    color="gray",
    fontsize=10,
    bbox=dict(
        boxstyle="round,pad=0.2",
        facecolor="white",
        edgecolor="none",
        alpha=0.8
    )
)

# Mark intersection points
points = [
    (tR_DS, A),
    (tE_DS, A),
    (tR_FXS, A),
    (tE_FXS, A)
]

for t_value, ae_value in points:
    ax.vlines(
        t_value,
        ymin=20,
        ymax=ae_value,
        color="gray",
        linewidth=1.0,
        linestyles=":",
        alpha=0.6
    )

    ax.plot(
        t_value,
        ae_value,
        "o",
        color="black",
```

```
        markersize=4
    )

# DS time gap
ax.add_patch(
    FancyArrowPatch(
        (tR_DS, 24.9),
        (tE_DS, 24.9),
        arrowstyle="<->",
        mutation_scale=10,
        linewidth=1.4
    )
)

ax.text(
    (tR_DS + tE_DS) / 2,
    25.45,
    f"DS gap = {H_DS:.2f} mo",
    ha="center",
    va="bottom",
    fontsize=10,
    bbox=dict(
        boxstyle="round,pad=0.25",
        facecolor="white",
        edgecolor="gray",
        alpha=0.95
    )
)

# FXS time gap
ax.add_patch(
    FancyArrowPatch(
        (tR_FXS, 22.9),
        (tE_FXS, 22.9),
        arrowstyle="<->",
        mutation_scale=10,
        linewidth=1.4
    )
)

ax.text(
    (tR_FXS + tE_FXS) / 2,
    22.35,
    f"FXS gap = {H_FXS:.2f} mo",
    ha="center",
    va="top",
    fontsize=10,
    bbox=dict(
        boxstyle="round,pad=0.25",
        facecolor="white",
        edgecolor="gray",
        alpha=0.95
    )
)

# Label the chronological ages
ax.text(tR_DS, 20.25, f"{tR_DS:.2f}",
        ha="center", va="bottom", fontsize=9)

ax.text(tR_FXS, 20.25, f"{tR_FXS:.2f}",
        ha="center", va="bottom", fontsize=9)

ax.text(tE_FXS, 20.25, f"{tE_FXS:.2f}",
        ha="center", va="bottom", fontsize=9)
```

```
ax.text(tE_DS, 20.25, f"{tE_DS:.2f}",
        ha="center", va="bottom", fontsize=9)

ax.set_xlim(41, 58.5)
ax.set_ylim(20, 28)

ax.set_xlabel("Chronological age (months)")
ax.set_ylabel("MSEL age-equivalent score (months)")

ax.grid(True, alpha=0.22)
ax.legend(loc="upper left", ncol=2, frameon=True)

fig.tight_layout()

fig.savefig(
    "figure2_time_gap_example.png",
    dpi=300,
    bbox_inches="tight"
)

plt.show()
```